

Deep Learning for Acoustic Insect Species Identification: A Comparative Study of Spectrogram-Based Architectures

**Synopsis report on
Deep Learning Project
[ICT-4442]**

Submitted By

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PART A : SYNOPSIS (Phase 1)

Name	Reg. No.	Model / Task Owned
Harsh Kumar Roy	230911296	Audio Spectrogram Transformer; temperature-conditioning contribution; common evaluation protocol and cross-model comparative analysis
Saurabh Tiwari	230911238	CNN and EfficientNetV2-S transfer learning; log-Mel spectrogram feature extraction; data augmentation pipeline and convolutional baseline tuning
Saksham Gupta	230911186	CRNN (convolutional front-end with BiLSTM); temporal modelling of stridulation pulse trains; class-imbalance strategies and long-tail performance analysis
Abhishek Patro	230911148	MLP baseline on handcrafted acoustic features; dataset acquisition and subset curation; preprocessing pipeline, resampling, chunking and split verification

Title of the Project: Deep Learning for Acoustic Insect Species Identification: A Comparative Study of Spectrogram-Based Architectures

Brief Description of the Project:

Insect populations are declining globally, yet monitoring coverage remains poor because conventional surveys require trained entomologists and are slow, costly and invasive. Many insect groups — notably Orthoptera (crickets, grasshoppers, katydids) and Cicadidae (cicadas) — produce loud, species-specific sounds, which makes passive acoustic monitoring an attractive alternative: microphones are inexpensive, deployable at scale, and record continuously without disturbing the habitat. The bottleneck is not recording but recognition.

This project builds and comparatively evaluates deep learning classifiers that identify insect species from short audio recordings. Audio is resampled, segmented into fixed-length chunks and converted into log-Mel spectrograms, reducing species identification to supervised classification over time–frequency images. Four models spanning distinct architecture families are implemented under a single shared evaluation protocol: a multilayer perceptron on handcrafted acoustic features (classical baseline), a convolutional network on log-Mel spectrograms, a convolutional–recurrent network adding a bidirectional LSTM to model the temporal structure of stridulation pulse trains, and an attention-based Audio Spectrogram Transformer. Identical splits and identical metrics isolate the effect of architectural inductive bias.

The primary dataset is InsectSet459, an openly licensed corpus of 26,298 recordings (226.6 hours) across 459 species. Its published benchmarks reach only about 57% macro-F1, so the task is genuinely unsolved rather than a saturated benchmark. Two properties make it demanding: a strongly long-tailed class distribution, in which roughly half of all species have fewer than 25 recordings, and highly variable recording conditions and sample rates arising from citizen-science collection.

Beyond the architectural comparison, the project investigates one underexploited signal. Insect stridulation is strongly temperature dependent — carrier frequency and pulse rate both shift with ambient temperature, documented since Dolbear's law in 1897. The dataset records ambient temperature and its authors identify this as potentially useful, but the published baselines do not use it. This project conditions the classifiers on temperature as an auxiliary input and measures the resulting change in accuracy.

The expected outcome is a reproducible comparison across four architectures on common metrics, an analysis of how performance degrades along the long tail, and a quantified assessment of whether temperature conditioning improves recognition.

Dataset(s) Identified:

Primary — InsectSet459 (Faiß, Ghani & Stowell, Scientific Data, 2026): 26,298 mono WAV/MP3 files, 226.6 hours, 459 species (Orthoptera 310, Cicadidae 149), curated from xeno-canto, iNaturalist and BioAcoustica. Sample rates 8–500 kHz (about one third ultrasonic). Weak labels, one species per file. Official 60/20/20 train/validation/test split. Strongly long-tailed: the most common species exceed 500 recordings while roughly half have fewer than 25. Metadata includes geolocation and ambient temperature (~19% of files). Licensed CC-BY-4.0 / CC0 via Zenodo (DOI 10.5281/zenodo.14056457).

Secondary — InsectSet66 (Faiß, 2023): 66 species, ~24 hours, same curation lineage; used for pipeline validation and comparison against published results. Working subset: main experiments use InsectSet459 species with at least 40 recordings, keeping training tractable within available compute; the full 459-class problem is retained as an extension. All four models use identical subsets and splits.

Proposed Models for Comparison:

Family	Model	Rationale
Classical / MLP	MLP on handcrafted features (MFCCs and deltas, spectral centroid, bandwidth, roll-off, ZCR)	Non-deep reference point; shows how much is solvable without learned features
CNN	Log-Mel CNN (VGG-style, plus EfficientNetV2-S transfer learning)	Standard bioacoustics approach; captures local time–frequency texture
RNN / Hybrid	CRNN — convolutional front-end with bidirectional LSTM	Insect calls are periodic pulse trains; tests whether explicit sequence modelling adds information
Attention	Audio Spectrogram Transformer, fine-tuned from AudioSet weights	State of the art for audio classification; global self-attention over the spectrogram

Model-to-Member Assignment: Harsh Kumar Roy (230911296), team lead — implements the Audio Spectrogram Transformer, designs and enforces the common evaluation protocol under which all four models are compared, conducts the temperature-conditioning experiments that constitute the project's principal technical contribution, and produces the cross-model comparison and error analysis. Saurabh Tiwari (230911238) — implements the convolutional arm of the comparison, covering both the VGG-style log-Mel CNN and EfficientNetV2-S transfer learning, and builds the waveform and spectrogram augmentation pipeline shared across all models. Saksham Gupta (230911186) — implements the convolutional–recurrent model, testing whether explicit sequence modelling of stridulation pulse trains adds information beyond convolution, and evaluates class-weighting and resampling strategies against the long-tailed species distribution. Abhishek Patro (230911148)

— implements the handcrafted-feature MLP that anchors the comparison, and is responsible for dataset acquisition, working-subset curation, and the preprocessing pipeline covering resampling, chunking and split verification.

Existing State-of-the-Art Methods Related to the Proposed Project:

- InsectEffNet (Faiß, Ghani & Stowell, 2026) — EfficientNetV2-S over 128-band Mel spectrograms in 5-second chunks with per-class weighting; ~56% macro-F1 and 72.2% accuracy on InsectSet459, and 95.6% F1 on the 66-species set.
- PaSST (Koutini et al., 2022; applied by Ghani et al., 2024) — patchout audio transformer and the strongest published InsectSet459 result at ~58% macro-F1; notably the CNN and transformer converge despite very different inductive biases.
- Adaptive representations (Faiß & Stowell, 2023) — the trainable LEAF frontend outperforms fixed Mel filterbanks on insect sound.
- BirdNET (Kahl et al., 2021) — 77.7% accuracy over 984 bird species; the gap insect acoustics has yet to close.

Published per-species results exceed 80% accuracy for well-represented species but decline sharply across the long tail, and both baselines discard information above 16–22 kHz despite a third of the corpus being ultrasonic. These limitations motivate the objectives below.

Objectives:

1. To establish a quantitative comparison of four deep learning architecture families — MLP, CNN, CRNN and Transformer — for acoustic insect species identification under a single common evaluation protocol.
2. To determine whether conditioning classifiers on ambient recording temperature improves species discrimination, given the known thermal dependence of insect stridulation.
3. To characterise the relationship between per-species training data volume and classification performance across the long-tailed species distribution.

Importance / Motivation of the Proposed Project:

Insects underpin pollination, decomposition and food webs, and reported population declines are a global ecological concern, yet assessment is limited by scarce monitoring data. Acoustic monitoring is minimally invasive, automatable and inexpensive at scale, but depends on reliable recognition. A recent systematic review found machine learning studies for insect sound recognition covered only about 302 species, and existing corpora are heavily biased towards Europe and North America, leaving tropical regions including India underrepresented despite far greater biodiversity. Bird classifiers are already standard monitoring tools while insect acoustics lags well behind — making this an unsaturated setting for architectural comparison, with direct relevance to pest surveillance and biodiversity assessment.

Preliminary Reference List:

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- [4] A. B. Kohlberg, C. R. Myers, and L. L. Figueroa, "From buzzes to bytes: a systematic review of automated bioacoustics models used to detect, classify and monitor insects," *Journal of Applied Ecology*, 2024.
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- [10] A. E. Dolbear, "The cricket as a thermometer," *The American Naturalist*, vol. 31, no. 371, pp. 970–971, 1897.

Work Plan:

Sl. No.	Component / Work Element / Milestone	Expected Completion (Day/Month)
1	Synopsis finalisation and submission	25 Aug
2	Dataset acquisition, working-subset curation, repository and environment setup	31 Aug
3	Preprocessing pipeline: resampling, chunking, log-Mel and handcrafted feature extraction, augmentation	07 Sep
4	MLP baseline and CNN implemented with preliminary results	16 Sep
5	Literature review table, contribution log, interim report; interim submission	25 Sep
6	CRNN and Audio Spectrogram Transformer implemented and evaluated	13 Oct
7	Temperature-conditioning and class-imbalance experiments; ablation study	20 Oct
8	Comparison table, error analysis, final presentation and report submission	24–31 Oct